Brute Force Attack: Simulation, Detection, and Defense

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1 Introduction

A **brute force attack** is a method where an attacker tries multiple password or key combinations repeatedly until the correct one is found. It is one of the simplest yet most common attack techniques used to gain unauthorized access to systems, applications, or accounts.
Disclaimer: The following content is for **educational and ethical purposes only**. These techniques should only be executed in a **controlled lab environment** with explicit permission. Unauthorized use against systems you do not own or manage is illegal and punishable by law.

2 Types of Brute Force Attacks

- **Simple brute force attack** – Directly attempting all possible combinations of passwords.
- **Dictionary attack** – Using a predefined list of likely passwords (a "dictionary") instead of trying every possible combination.
- **Hybrid brute force attack** – Combining dictionary attacks with variations (e.g., adding numbers or symbols).
- **Reverse brute force attack** – Starting with a single commonly used password and trying it against many usernames.
- **Credential stuffing** – Using leaked username-password pairs from previous data breaches to attempt logins on other systems.

3 Popular Brute Force Tools

- **John the Ripper** – A fast password cracker for Unix, Windows, and other platforms.
- **Hashcat** – A powerful GPU-based password recovery tool, ideal for cracking password hashes.
- **Aircrack-ng** – Focused on cracking Wi-Fi network keys.
- **L0phtCrack** – Used to audit and recover Windows passwords.

In cybersecurity, understanding how attackers operate is essential for building **robust defenses**. Security analysts often use these tools in penetration testing to identify weaknesses before malicious actors exploit them.

4 Hydra for Brute Force Attacks

Hydra is one of the most widely used tools for network login brute forcing. It can perform rapid dictionary-based attacks against more than 50 protocols, including **HTTP, SSH, FTP, RDP, Telnet**, and more.

4.1 Hydra Command Structure

```
hydra -L <username_file> -P <password_file> <target>
```

- `-L` : Specifies the file containing a list of usernames.
- `-P` : Specifies the file containing a list of passwords.

5 Brute Force SSH

A brute force SSH attack attempts to gain unauthorized access to a remote machine by repeatedly trying different username and password combinations against the **SSH service (port 22 by default)**.

5.1 Example Hydra Command

```
hydra -L <username_file> -P <password_file> ssh://<target_ip>
```

To run this attack, you need:

- Target IP address
- The service to attack (**SSH**)
- The login port number (default: **22**)
- A username list and a password list to test

6 System Requirements

- **Attacker Machine**: Kali Linux
- **Target Machine**: Windows system (with SSH enabled)

  Note: Since Windows does not enable SSH by default, you must install and enable OpenSSH Server before running this test.

7 Simulating an SSH Brute Force Attack with Hydra

7.1 Step 1: Preparing the Password List

On the **attacker machine (Kali Linux)**, we first create a list of possible passwords in a file named `passwords.txt`. This file will be used by Hydra to attempt login combinations against the target system.

Figure 1: Password file used for brute force simulation

7.2 Step 2: Running the Hydra Command

Since we already know the username of the target system is `dell`, we use the `-l` option to specify it directly. If we did not know the username, we could use the `-L` option along with a file containing a list of usernames.

```
hydra -l dell -P passwords.txt ssh://<target_ip>
```

- `-l dell` → specifies the username directly.
- `-P passwords.txt` → specifies the file containing possible passwords.
- `ssh://<target_ip>` → specifies the target machine's IP address and service.

Figure 2: Running Hydra against SSH service

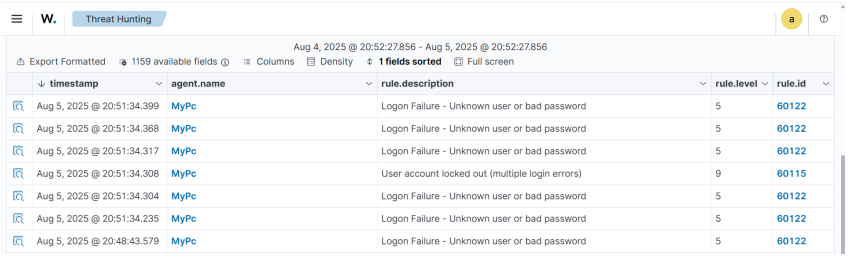
7.3 Step 3: Observing the Attack

Hydra will automatically attempt multiple login attempts using the given username and password list. Each failed login attempt is logged by the target's SSH service.

8 Detection in Wazuh

After simulating the brute force attack, we switch to the **Wazuh dashboard**. In the **Threat Hunting** section, we can see the following indicators:

- Multiple **failed login attempts** for the SSH service.
- Alerts triggered by **Wazuh security rules** detecting brute force activity.
- Event details, such as the source IP (attacker machine) and the affected service (SSH).



Threat Hunting

Aug 4, 2025 @ 20:52:27.856 - Aug 5, 2025 @ 20:52:27.856

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timestamp	agent.name	rule.description	rule.level	rule.id
Aug 5, 2025 @ 20:51:34.399	MyPc	Logon Failure - Unknown user or bad password	5	60122
Aug 5, 2025 @ 20:51:34.368	MyPc	Logon Failure - Unknown user or bad password	5	60122
Aug 5, 2025 @ 20:51:34.317	MyPc	Logon Failure - Unknown user or bad password	5	60122
Aug 5, 2025 @ 20:51:34.308	MyPc	User account locked out (multiple login errors)	9	60115
Aug 5, 2025 @ 20:51:34.304	MyPc	Logon Failure - Unknown user or bad password	5	60122
Aug 5, 2025 @ 20:51:34.235	MyPc	Logon Failure - Unknown user or bad password	5	60122
Aug 5, 2025 @ 20:48:43.579	MyPc	Logon Failure - Unknown user or bad password	5	60122

Figure 3: Wazuh detecting brute force attack

9 Defensive Countermeasures

While detection is critical, **prevention** strengthens system defenses against brute force attempts. Recommended strategies include:

- Use strong and unique passwords.
- Enable multi-factor authentication (MFA).
- Regularly monitor login activity.
- Implement rate-limiting to restrict login attempts and lock accounts after repeated failures.

10 Conclusion

Brute force attacks remain one of the most common methods used by attackers to gain unauthorized access to systems and services. In this report, we demonstrated how such an attack can be simulated using the Hydra tool against an SSH service. We also observed how Wazuh is capable of detecting these attacks in real time, highlighting multiple failed login attempts in the dashboard.

This exercise shows the importance of monitoring and detection tools like Wazuh in strengthening system defenses. While brute force attacks are relatively simple, they can still be effective if strong passwords and proper security measures are not in place. In future work, deeper analysis of Wazuh logs and alerting mechanisms will provide further insights into identifying, investigating, and mitigating brute force threats.